Skills Summary

Choosing and Combining Factoring Methods

Use this reference while completing your worksheet or when studying.

Run this checklist in order on every expression.

- **1. Common factor?** Pull out the GCF of all terms, variables included. Then look at the bracket and start again from step 2.
- **2. How many terms?** Two terms with a minus sign: difference of squares, $A^2 - B^2 = (A-B)(A+B)$. Three terms: trinomial.
- **3. Perfect square?** $A^2 \pm 2AB + B^2 = (A \pm B)^2$ — square ends, middle term equal to $2AB$.
- **4. Leading coefficient 1 or not?** If 1, find a factor pair of the constant adding to the middle coefficient. If not, group on the product of the outer coefficients.
- **5. Factor again.** Every bracket gets the checklist too. A sum of squares, $A^2 + B^2$, is where you stop.

1. Common factor first, then look again

The GCF step is not optional politeness — it is what makes the later pattern visible. A coefficient that is not a perfect square ($2x^2$, $3x^2$) or a term of degree three or higher is the usual sign that a common factor is hiding a pattern underneath.

Example: Factor $2x^3 - 8x$ completely.
Step 1. Two terms and a minus sign look like a difference of squares, but $2x^3$ is not a square — so run checklist step 1 first and take the common factor out.
Step 2. Both terms divide by 2 and both carry an x, so the GCF is $2x$: $2x^3 - 8x = 2x(x^2 - 4)$, and $x^2 - 4 = (x-2)(x+2)$. $2x^3 - 8x = \mathbf{2x(x-2)(x+2)}$

Try it: Factor $3x^3 - 27x$ completely.

check: $\mathbf{3x(x-3)(x+3)}$

2. Trinomials with leading coefficient 1

$x^2 + bx + c = (x+p)(x+q)$ where $p + q = b$ and $pq = c$. Read the signs before hunting: if $c > 0$ both numbers share the sign of b; if $c < 0$ they have opposite signs, and the bigger one carries the sign of b.

Example: Factor $x^2 + 7x + 12$.
Step 1. No common factor and the leading coefficient is 1, so this is the factor-pair case — no grouping needed.
Step 2. Need a pair multiplying to 12 and adding to 7: 3 and 4, since $3 \cdot 4 = 12$ and $3 + 4 = 7$. $x^2 + 7x + 12 = \mathbf{(x+3)(x+4)}$

Try it: Factor $x^2 - 2x - 15$.

check: $\boldsymbol{(x-5)(x+3)}$

3. Trinomials with a leading coefficient that is not 1

For $ax^2 + bx + c$ with $a \neq 1$: find a factor pair of the product ac that adds to b, split the middle term into those two pieces, then factor by grouping in pairs. The two brackets you get must match, which is the built-in check that the split was right.

Example: Factor $3x^2 + 11x + 6$.
Step 1. No common factor and the leading coefficient is 3, so factor pairs of the constant alone will not work — group on the product ac instead.
Step 2. $ac = 3 \cdot 6 = 18$, and $9 + 2 = 11$, so split: $3x^2 + 9x + 2x + 6 = 3x(x+3) + 2(x+3)$.
$3x^2 + 11x + 6 = \boldsymbol{(3x+2)(x+3)}$

Try it: Factor $2x^2 - 5x - 3$.

check: $\boldsymbol{(2x+1)(x-3)}$

4. Special forms: squares you can spot

$A^2 - B^2 = (A-B)(A+B)$ and $A^2 \pm 2AB + B^2 = (A \pm B)^2$. Squares to know on sight: $4, 9, 16, 25, 36, 49, 64, 81, 100$; also $4x^2 = (2x)^2$, $9x^2 = (3x)^2$, $x^4 = (x^2)^2$. Recognising a special form skips all trial and error.

Example: Factor $4x^2 - 20x + 25$.
Step 1. Three terms whose first and last are perfect squares — test the perfect-square pattern before grouping, because it is far quicker when it works.
Step 2. $A = 2x$ and $B = 5$; the pattern needs a middle term of $2AB = 2 \cdot 2x \cdot 5 = 20x$, and the trinomial has $-20x$, so the bracket is a difference. $4x^2 - 20x + 25 = \boldsymbol{(2x-5)^2}$

Try it: Factor $9x^2 - 24x + 16$.

check: $\boldsymbol{(3x-4)^2}$

(!) Watch out: A correct product is not always a complete factorization. $(2x+4)(x+2)$ really does multiply back to $2x^2 + 8x + 8$, but $2x + 4$ still hides a common factor of 2 — the complete answer is $2(x+2)^2$. Taking the GCF out first would have prevented it.